SSHVisual: Visual Tool to Analyze SSH Data

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***Abstract*—Network operators use visualizations to detect attacks on their systems, interacting with data to uncover patterns that might stay hidden. Operators use these findings to block attackers and to expand the knowledge base of known threats. Detecting anomalies and identifying relationships iteratively in data is, at best, tedious and, at worst, infeasible without data visualizations. Prior work in network visualization analyzed the IP address and ASN dimensions of SSH logins with little focus on the username aspect of the attackers' attempts and detected anomalous network events when baseline events were available. As a result, little attention has been paid to whether attackers that differ in timing, source address, and username belong to the same coordinated campaign. We present SSHVisual, an iterative visualization system evaluated on login attempts collected from our production SSH systems. SSHVisual analyzes the IP, ASN, and username dimensions of SSH attacks, to support campaign detection. It complements automated detection by allowing operators to find anomalous events without a normal baseline. It uses a similarity search across dimensions that groups attackers by behavior and explains each match. It supports iterative and collaborative sharing of findings between operators. We evaluated SSHVisual on 206.6 million attacker login attempts collected over 69 days. We found that it helps operators surface hidden patterns and find attackers to block. In one case study, some IPs tried a similar set of usernames, including uncommon ones, which pointed to a single obscure campaign that volume charts could present as unrelated bursts. In another, similarity search surfaced low-volume IPs working through various username lists, none of which ranked in the top ten by volume. Without similarity search, these IPs would likely have been hard to find and block. These results suggest that username-centric, operator-driven exploration can be valuable because it can reveal covertness and coordination that previous analyses can miss.**

***Index Terms*—SSH, brute-force attacks, attack campaigns, network security visualization, visual analytics, similarity search**

I. Introduction

Computer systems employ SSH [1] to allow users to securely connect to them in order to execute commands and do remote administration. Many users do a username, password combination (password based authentication) in order to gain entry to the SSH system. Attackers, who are adversarial, can use brute force attack mechanisms [2] to get into SSH systems. Sometimes, they guess the username, password combinations and other times, they take advantage of poorly secured systems to build botnets [2] to carry out further attacks [3], [4] or steal resources from the systems, like cryptomining [5], [6].

Given the prevalence of SSH attacks in the IPv4 space [2], we designed SSHVisual and evaluated it on login attempts from our production SSH systems. SSHVisual serves two kinds of network operators: Network administrators, who handle daily SSH traffic, need to identify attackers quickly and block them and Security analysts, who study traffic over longer periods, look for patterns that are not apparent and share what they find. Consider a single period: an administrator can isolate the highest-volume attacker and block it, while an analyst examining the same window can iteratively surface a set of low-frequency attackers that attempt an identical sequence of usernames, and group them as a coordinated campaign.

Surveys of network visualization [7]–[11] establish the maturity of the field and the gaps we target. Our two contributions follow from treating the username, alongside the source IP and ASN, as part of an attacker's identity, and each one serves one of the two operators described above. First, for security analysts, we show that username-based similarity identifies coordinated campaigns that volume-based views split apart. IPs that try the same list of usernames can be recognized as a single campaign, even when they come from different countries and ASNs and are never active on the same day. Second, for network administrators, we show that behavioral similarity search finds attackers that would otherwise stay hidden. Starting from a single IP the administrator already intends to block, it surfaces low-volume IPs that act alike but never reach the top-ten volume rankings, and it explains each match, so the administrator knows which other IPs to block and why. Since network traffic may not offer a stable, attack-free baseline of normal behaviour [12], detectors that rely on such a baseline may miss activity that doesn't deviate from it in the expected way. Operators, by contrast, may recognize coordination and similarity across sessions visually, without specifying normality in advance. The above contributions build on this ability and are meant to complement baseline-driven detection. As operators explore, they can label and share findings so colleagues do not investigate the same patterns twice, and the findings can in turn inform automated detection.

The rest of the paper is as follows: Section II is Related Work, Section III is Tasks and Requirements, Section IV is Design, Section V is Evaluation and Section VI is Conclusion.

II. Related Work

A long line of visualization tools [13]–[22] renders network traffic and security events navigable for operators, the majority being network-flow and traffic oriented [13], [14], [16],

[18], [19], [21], [22], with some also integrating intrusion-detection/prevention and host logs [15], [17] or multidimensional cyber-defense data [20]. Additionally, [23] visualizes IDS alerts and [24], [25] show heterogeneous security logs. [26] extract multi-resolution discrete-wavelet features from traffic, project them with principal component analysis, and apply hierarchical clustering across several intrusion-detection benchmarks, separating brute-force attacks into distinct clusters. Related work on coordinated multiple views [27]–[29] and insight generation [30] informs our design.

A. Username Campaigns with Time

To our knowledge, honeypot literature has rarely treated the username identity as an actor for campaign detection purposes. Typically, it is a dimension to count rather than to track, whereas in our analysis it serves, alongside the IP and ASN, as an anchoring dimension. Several papers [31]–[37] collect or chart honeypot traffic, with several of them capturing SSH credentials (usernames and passwords). Where credentials are charted, like the SCADA SSH honeypot [34], kippo-graph [35], and the container deployment [36], usernames and passwords appear as static top-N tallies (kippo-graph's top-10 username and password bars being an example), and not placed on a time axis. The WAN honeypot framework [38] is the closest prior system : it computes an IP-to-IP dictionary-overlap (containment) score to probe whether six same-/24 source IPs share a wordlist, which is a manual, one-off prototype of credential-based coordination analysis, performed on the IP and without a temporal dimension.

B. Username-Based Similarity

Computed similarity between entities is well established in literature [39]–[42]. We do not claim similarity computation itself as novel; the contribution lies in the entity it operates on. Prior systems cluster internal hosts [39], behavioral profiles [41], IP/port members [42], or internal enterprise users grouped by shared resources [40]. None treat the username as an attacker entity. We use the username as a signature of an IP, grouping IPs by the credentials they try, with uncommon usernames weighted as stronger evidence than ones nearly every attacker tries. Honeypot/honeynet analyses have related attack activity through time by grouping attacks whose time series rise and fall together [43], [44]. Attackers that share a credential list, however, need not be active together, so we compute similarity independently of when each attacker is active and let the operator see their time series in a linechart.

C. Limits of Automated Detection

Automated anomaly detection assumes a normal baseline, and SSH traffic generally does not have one. Even where a baseline exists, such detection is fragile. [45] find that machine-learning anomaly detection is rarely deployed, because normal traffic varies widely, errors are costly, and sound evaluation is hard. [46] show that when intrusions are rare, the false-alarm rate limits any intrusion detector. Our data is the opposite case, since nearly every login attempt is an attack. [12] question whether attacks are rare and whether attack-free training data exists at all, and note that malicious traffic can look normal. [39] and [47] learn a model of each actor's expected behavior with [47] assuming that an actor's activity repeats weekly. However, constantly churning, near-uniformly hostile traffic like we see cannot offer a per-actor normal. Situ also concedes that operators "do not want another automated tool with algorithms they do not trust" and keeps a human in the loop. Lacking a per-actor baseline, we make campaign identification operator-driven from the start.

D. Username as Actor Identity

We analyze attackers at the level of a campaign, meaning a group of attackers acting together, rather than one attacker at a time. That raises a practical problem: a campaign rarely shows up as a single source IP. IPs may not be tied to one machine. One IP can be assigned to different hosts [48], hosts can cycle through new IPs [49], and counting by address can overstate the number of real machines sometimes by an order of a magnitude [50]. A view organized by address is therefore likely to scatter one campaign across dozens of entries that can look unrelated. The username can pull those pieces back together. Attackers in the same campaign likely draw from a shared credential list, so the same login names may recur across changing IPs. Grouping by username can reassemble the campaign that an address-by-address view could split apart.

E. Operator Labeling and Collaboration

The findings an operator produces can be captured and shared. In two works [19], [51], interactive labeling was used to fold the operator's judgments back into the system. Collaborative analysis was supported through shared artifacts with [17] and [20] being examples. These systems establish that labeling and collaboration are valuable, which is why our labeling-and-sharing mechanism is the step that turns an identified set of usernames into a durable, shared campaign.

III. Tasks and Requirements

The design of SSHVisual draws on our experience as network operators and on gaps we identified in prior work on SSH attack analysis and security visualization. From these we derive three tasks and five requirements.

A. Tasks

T1: Identify SSH campaigns. Security analysts need to find attackers that behave alike across usernames, IPs, ASNs, and countries, and record groups so they can be referenced.

T2: Detect anomalies that warrant action. Network administrators need to see unusual activity, like a sudden spike or a source dominating traffic, and decide whether to block it.

T3: Share findings across operators. Analysts and administrators need to see what their colleagues have already investigated, so that the same patterns are not identified twice.

B. Requirements

R1: Attacker similarity profiling. SSHVisual should be able to cluster similar attackers along the username, IP and ASN dimensions. R1 mostly serves T1 as it allows security analysts to build a repository of similar attackers.

R2: Sorting and ranking. SSH data can be heavy-tailed: one username can account for a quarter of attempts (remaining attacks spread across other usernames) [52] and persistent IPs can account for most attempts [53]. SSHVisual should therefore list every entity in each dimension, and let the operator sort and rank entities by any combination of attributes. R2 supports T1 and T2: network administrators can surface anomalous entities that may not reach the top of a volume ranking, and security analysts can find similar entities.

R3: Shared tagging. Operators should be able to tag attackers, record what a tag means, and why it was applied to a particular attacker, and have those tags remain visible to every operator. Each tag application should record who applied it and when, so later operators can judge whether a finding is still current. R3 primarily supports T3, and supports T1 because tags accumulate into a repository of related attackers.

R4: Complementary tabular and temporal views. No single visualization can serve every need. A table can support precise lookup and ranking but can discard temporal structure; a linechart can preserve temporal structure, showing spikes and campaigns at a glance, but can convey values less precisely. SSHVisual should therefore provide both views and let operators move between them. A network administrator may find and act on an anomaly using the table alone, whereas a security analyst may use the table to assemble related attackers and examine their behavior over time in the linecharts. R4 supports T1 and T2.

R5: Interactive exploration of the linecharts. The linecharts follow Shneiderman's Visual Information-Seeking Mantra: overview first, zoom and filter, then details-on-demand [54]. They open on the full time range, let operators zoom into a time window and filter entities, and reveal an entity's details on request. R5 supports T1 and T2: zooming can isolate the time window of a suspected campaign, and filtering can remove dominant entities that would mask smaller ones. Line charts can become harder to read accurately as more series are added [55]. Each linechart should therefore show few entities and we decided to cap it at 10.

IV. Design

SSHVisual combines two visual components, a discovery table and a set of linecharts, with two non-visual components, similarity search and tagging. We describe the visual design in Section IV-A and the non-visual design in Section IV-B. All figures show SSH login attempts collected between 1 November 2022 and 8 January 2023, a period of 69 days.

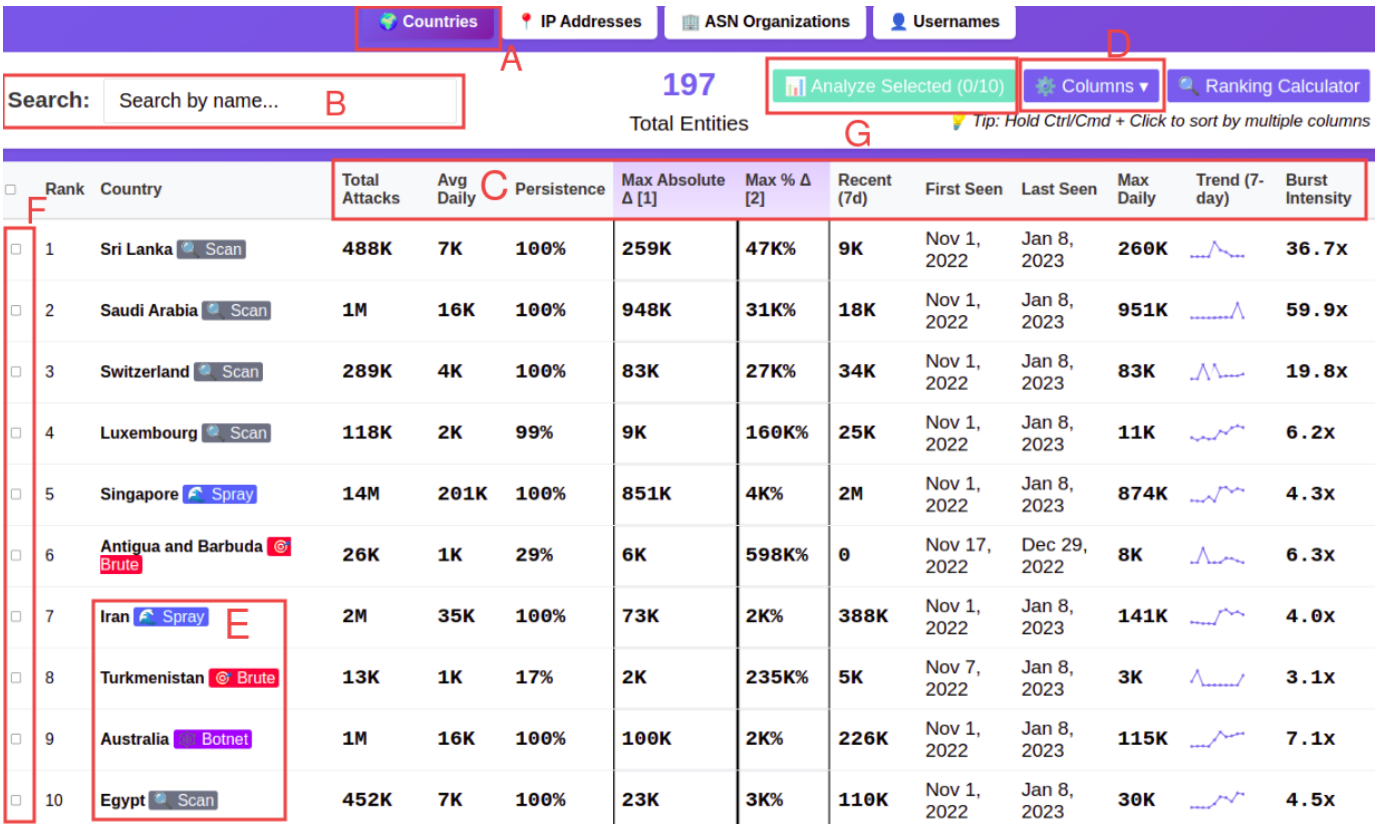


Fig. 1. Discovery table, showing the countries tab (A). Operators can filter countries(B). A row shows a country and operators can sort by any mix of columns (C), with more columns present in (D). Based on attack patterns, a badge is given (E). Operators can select up to 10 rows (F and G) to plot in linechart.

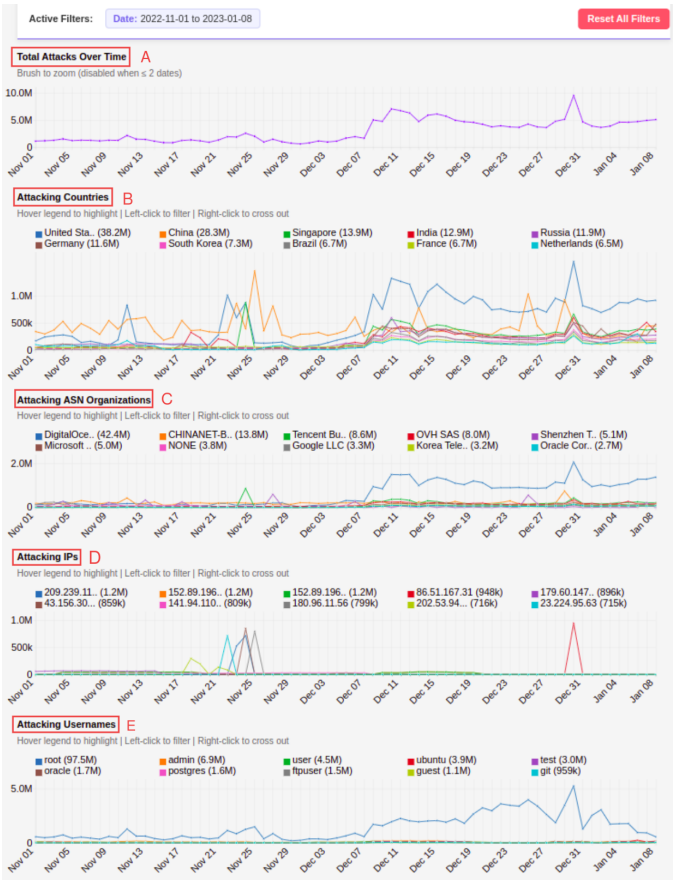


Fig. 2. The first chart (A) shows total attacks; the four charts below show the 10 most active Countries (B), ASNs (C), IPs (D), and Usernames (E).

A. Visual Design

The interface provides two views of the data. The discovery table (Fig. 1) is time independent: each row summarizes one entity over time. The linecharts (Fig. 2) can be time-dependent. Both views cover the full data period by default.

1) Discovery Table: The discovery table has four tabs, one per dimension: Countries, IP addresses, ASN organizations, and Usernames. In Fig. 1, the Countries tab is active (label A). A search box filters rows (label B), and the counter above the table reports the number of entities (197 countries). Some columns (label C) are shared across all tabs: total attacks, average daily attacks, persistence (the share of days on which the entity was active), the largest absolute and percentage daily change, attacks in the most recent seven days, first- and last-seen dates, maximum daily count, trend (line chart showing attacks at seven day intervals) and burst intensity (ratio of max daily attacks to average daily attacks where a higher ratio likely indicates an entity that attacks sporadically but with higher frequencies). The full list and figures of columns (labels C and D) are found in the Appendix, Section VI-A.

Exclusive to the countries tab, each country also carries a badge summarizing its dominant behavior, namely Scan, Spray, Botnet or Brute (label E). Brute marks entities whose traffic comes from few IP addresses making many attempts on few usernames. Spray marks traffic from many IP addresses trying many usernames, and Botnet marks traffic spread across many IP addresses and many ASNs. Scan marks a general attack pattern that fits none of the other three.

Unlike the linecharts, which show only the 10 most active entities per dimension, the table lists every entity. Operators can select up to 10 rows with the checkboxes (label F) and use Analyze Selected (label G) to plot those entities in the linecharts, and they can return to the linecharts at any time. This link lets operators move from a precise, ranked lookup to the temporal behavior of the same entities, fulfilling R4.

2) Linecharts: The linechart view (Fig. 2) stacks five charts on a shared time axis. The first, Total Attacks Over Time, shows the aggregate volume of attempts (label A); the remaining four show the 10 most active countries, ASNs, IPs, and usernames (labels B-E). Each legend entry lists the entity's total for the selected period. Interaction with the linecharts follows [54] (fulfilling R5), and the charts are coordinated so that an action in any one of them updates all of them.

Overview first. The charts cover all 69 days of data, with each chart showing the 10 most active entities in its dimension.

Zoom. Every chart supports brushing. When the operator brushes a time range on any chart, the range is applied to all five charts and recorded in the Active Filters bar, so the temporal context stays the same whichever dimension the operator is examining. Brushing is disabled once the range is two days or fewer, since a line needs more datapoints.

Filter. Right-clicking a legend entry crosses the entity out, removing it from its chart and rescaling the y-axis so that smaller series are no longer flattened against the axis by a dominant one. Left-clicking a legend entry selects that entity and filters every chart to the attacks associated with it. Because each attack attempt links a username, an IP, an ASN, and a country, the selection propagates along these relationships:

- Selecting a country restricts the ASN, IP, and username charts to the entities observed in attacks from that country
- Selecting an ASN shows the country it originates from, along with its IPs and the usernames they try
- Selecting an IP shows its country and ASN, along with the usernames it tries
- Selecting a username shows the countries, ASNs, and IPs from which it was tried

Details-on-demand. Hovering over a data point shows a tooltip with that entity's attack count for the day, and hovering over a legend entry highlights the entity's line, and the legend reports each entity's total for the selected period.

Linked filtering can turn the linecharts from four separate rankings into a means of tracing an entity across dimensions. An operator investigating the spike on 30 December, for example, can select `root` in the username chart; the IP, ASN, and country charts then show which addresses, networks, and countries tried `root`, and brushing around 30 December narrows the view to the spike itself. Conversely, selecting an ASN such as DigitalOcean reveals which usernames its hosts try, indicating whether traffic from a hosting provider can

	Rank	Username	Total Attacks	Avg Daily	Persistence	Max Absolute Δ [1]	Max % Δ [2]	Recent (7d)	First Seen	Last Seen	Max Daily
☐	1	karaf	77.6K	5.2K	21.70% (15d)	32.4K	360.52K%	120	Nov 2, 2022	Jan 8, 2023	44.8K
☐	2	ossuser	82.6K	2.6K	46.40% (32d)	30.4K	276.15K%	690	Nov 3, 2022	Jan 8, 2023	46.0K
☐	3	snow	248.2K	5.2K	69.60% (48d)	22.0K	168.87K%	41.8K	Nov 1, 2022	Jan 8, 2023	35.2K
☐	4	nmrsu	89.4K	1.8K	72.50% (50d)	35.0K	145.91K%	2.9K	Nov 5, 2022	Jan 8, 2023	43.8K
☐	5	lenovo	25.3K	476.51	76.80% (53d)	8.5K	283.10K%	5.7K	Nov 1, 2022	Jan 8, 2023	8.5K
☐	6	fb	9.7K	225.67	62.30% (43d)	4.7K	470.10K%	403	Nov 5, 2022	Jan 8, 2023	4.7K
☐	7	inspur	40.4K	684.71	85.50% (59d)	5.3K	263.05K%	11.2K	Nov 1, 2022	Jan 8, 2023	5.3K
☐	8	radware	12.3K	686.11	26.10% (18d)	4.8K	241.25K%	1.9K	Nov 1, 2022	Jan 8, 2023	4.8K

Fig. 3. Usernames tab sorted by Max Absolute Δ (A) and Max % Δ (B), surfacing bursty usernames (C).

be one campaign or many. Brushing on every chart, linked filtering, and on-demand details together fulfill R5.

The linecharts answer questions that the discovery table cannot, such as "Do the countries in Fig. 1 share the same peaks and valleys?" or "Does one country begin attacking earlier than another?" In Fig. 2, the aggregate chart shows a sharp spike on 30 December, and most of the top ten countries peak on the same day, indicating a coordinated, multi-country event rather than activity from a single source. The charts also reveal differences in timing. China's largest peak falls in late November, weeks before the US' activity rises, and the IP chart shows individual addresses that each spike for a day or two and then fall silent. Neither pattern can be recovered from the table's per-entity totals. Together, Figs. 1 and 2 show how the two views complement each other, fulfilling R4.

3) Sorting and Ranking: Fig. 3 shows the usernames tab sorted on two columns (labels A and B). Operators sort by holding Ctrl/Cmd while clicking headers, and the bracketed indices in the headers show when the column was clicked.

Sorting matters because the data are heavy-tailed. The username linechart in Fig. 2 is dominated by `root`, which accounts for tens of millions of attempts and flattens every other line. Ranking by day-over-day change instead surfaces usernames (label C in Fig. 3), which are not seen in Fig. 2, several of which are default accounts for specific software or hardware products (`karaf`, for example, is the default account of Apache Karaf). Their totals are far below the top ten, but their activity is concentrated: `karaf` received 77.6K attempts in total, 44.8K of them on a single day, and was active on only 15 of the 69 days. Bursts of product-specific credentials like these suggest campaigns targeting particular devices, yet they can be invisible in a volume-ranked chart. Fig. 3 thus shows how sorting and ranking over the full entity list fulfills R2.

B. Grouping and Tagging

The visual components help operators find individual attackers, while the grouping and tagging components help them connect those attackers to others and record what they find. Similarity search groups attackers by behavior, in the username dimension (Section IV-B1) and the IP dimension (Section IV-B2), and shared tagging (Section IV-B3) preserves the resulting groups so that other operators can build on them.

1) Username Similarity: Similarity search lets operators start from one attacker and retrieve the attackers that behave most like it. For usernames, the unit of analysis is the population of sources that tries a username. For a username, we compute the breadth of that population (the number of distinct IPs, ASNs, and countries that try it), its volume, traffic concentration on its most active source, among other features. We do not reuse the 17-dimensional profile we apply to IPs (Section IV-B2), because a diagnostic showed some features to be uninformative for usernames. Daily stability is an example: most IPs are active many days, so their stability is well defined, but half of the usernames appear only one day, where stability is meaningless. We cluster username profiles with k-means ($k = 9$). Each cluster is labeled automatically from its centroid. To answer a query, the system retrieves the query's cluster and ranks the other members by Euclidean distance to the query, reported as a similarity score.

In Fig. 4, an operator searches for `james` (label A). The system places it in Cluster 3, labeled High-Volume Persistent Botnet-Scale, which contains 2,624 usernames; a cluster member receives about thirty one thousand attempts from roughly four thousand IPs and is active on 72% of days (label B). The results table lists the 20 most similar usernames (five shown here), each with its total attacks, top country, top three source IPs, the top three ASNs, and a set of "Why similar?" labels (label D), such as a shared lead source country, comparable volume, or sourcing volume, that explain the match.

Above the results, a shared infrastructure panel (label C) lists IPs that appear among the top sources of more than one result, and these IPs are shown in the table. In Fig. 4, 37 IPs are shared. Each line of the panel names an IP and counts how many of the 100 results it is a top source for, followed by examples of those usernames. Two of the most shared addresses, 152.89.196.123 and 152.89.196.220, lie in the same /24 network and are top sources for 34 and 33 results respectively, including `manager`, `daniel`, and `oscar`. The panel applies the same analysis to ASNs: 13 are shared, led by DigitalOcean (99 of 100 results), Tencent (90), and OVH (40). This panel can turn behavioral resemblance into infrastructural evidence: usernames tried by the same hosts are likely drawn from the same credential list and belong to the same campaign.

2) IP Address Similarity: We characterize each IP address by a 17-dimensional profile drawn entirely from its activity during the data period. The features fall into four groups: volume (log-scaled totals, daily means, and peaks); temporal shape (the fraction of days on which the address was active,

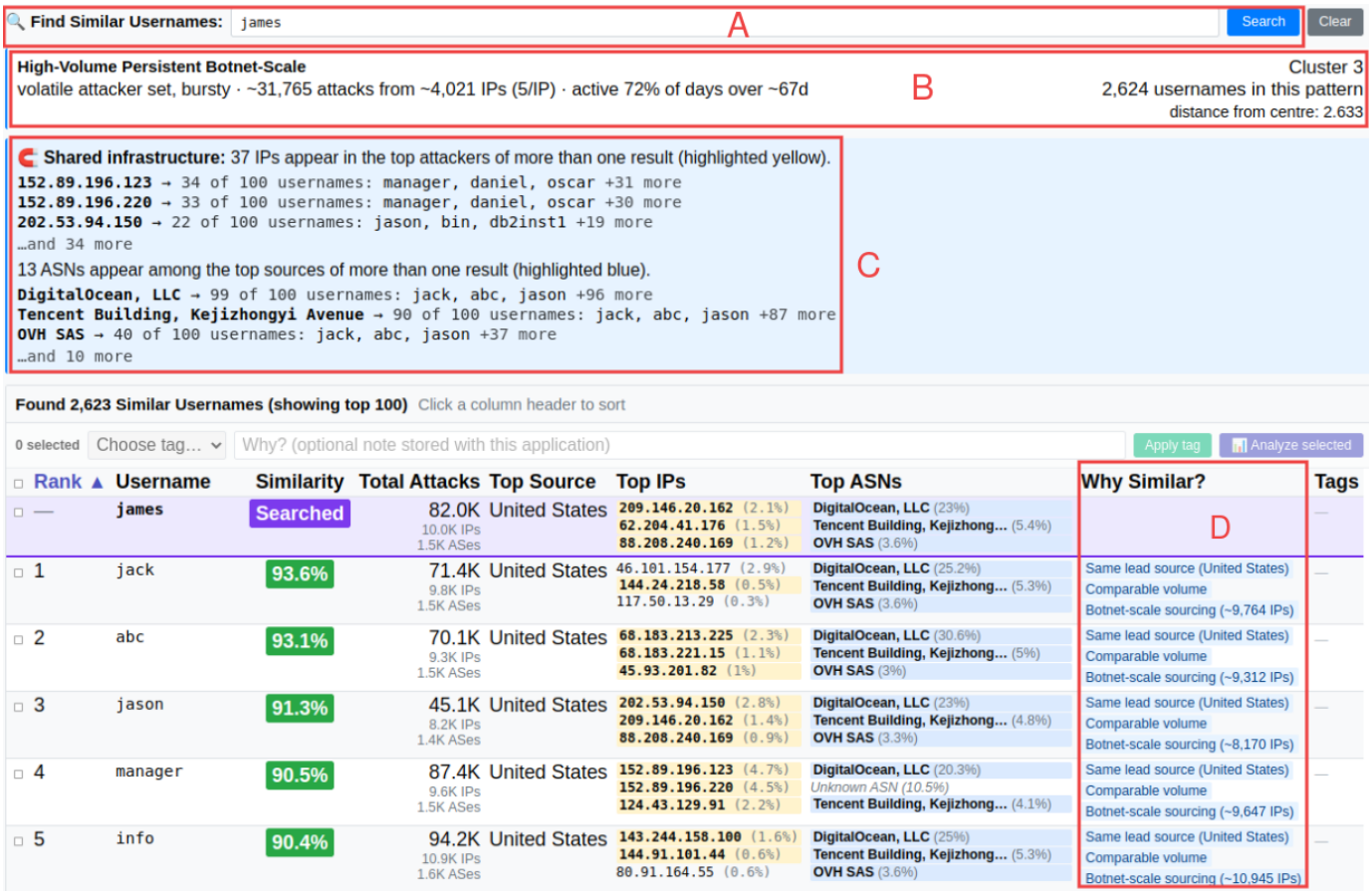


Fig. 4. Similarity search for `james` (A). Cluster summary (B), Shared infrastructure (C), and Why similar? (D) explain why the results were grouped.

its burst intensity, its largest daily swings, a fitted trend slope, its activity span, and the share of its traffic in the final week); targeting behavior (unique usernames tried, how stable that set is from one day to the next, its rotation rate, and the share of attempts taken by its most-tried username); and two flags indicating whether the address belongs to a cloud provider and whether it originates from a major source country. We cluster these profiles with k-means, label each cluster from its centroid, and answer queries in the same way as for usernames.

In Fig. 5, an operator searches for 61.177.173.46 (label A), which falls in Cluster 5, labeled Low-Volume Persistent Exploratory, containing 4,405 addresses. The most similar results are shown in label B, with similarity scores between 91.2% and 80.4%, all belonging to CHINANET-BACKBONE. Behavioral features alone can recover infrastructure relationships, suggesting that these hosts may be operated together.

Figs. 4 and 5 show how SSHVisual lets operators assemble sets of similar attackers along the username and IP dimensions and see why those attackers were grouped, fulfilling R1.

3) Shared Tagging: Figs. 6–8 show how an operator records a finding so that other operators can see work that has already been done. Tags are organized in a tag repository for each dimension. The repository displays the current operator's name, and every tag and tag application records its author.

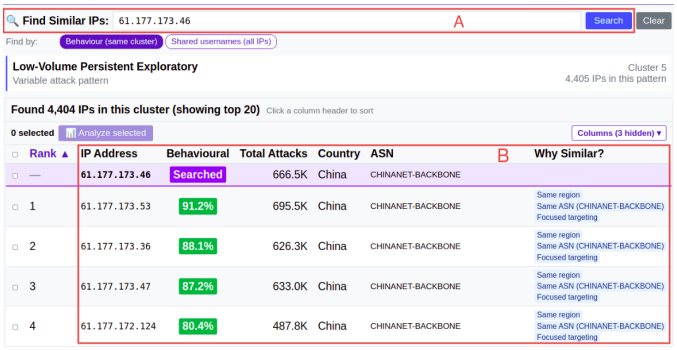


Fig. 5. Similarity search (A) for IP 61.177.173.46. Nearest neighbors are adjacent addresses on the same network (B).

In Fig. 6 the operator creates a tag, "similarity to james" (label A), for the username cluster found in Fig. 4. The repository lists the tag with its description (label B) and its author (label C). Tags are applied from the similarity results. In Fig. 7 the operator selects seven usernames (label A), chooses the new tag, and adds a note explaining this particular application: "recurring ASNs" (label B). In Fig. 8 the repository reports seven applications (labels A and F). Filtering the list by tag shows each tagged username with its note (labels B and

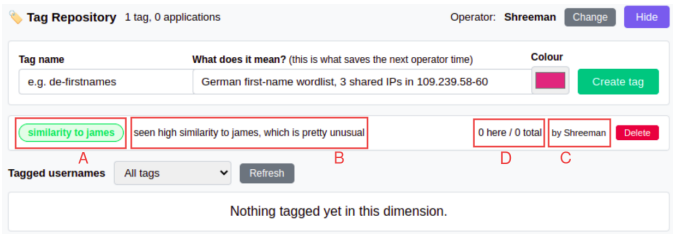


Fig. 6. A newly created tag (A), listed with its description (B), author (C), and application counts (D).

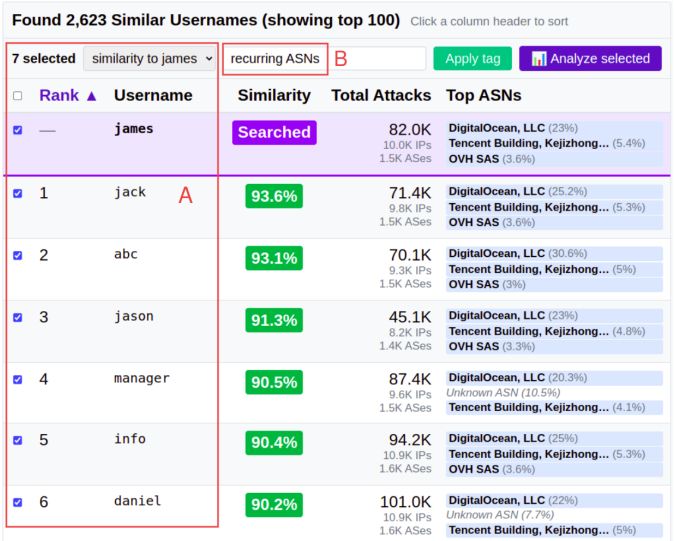


Fig. 7. Applying the tag to seven usernames from the similarity results (A), with a note explaining the application (B).

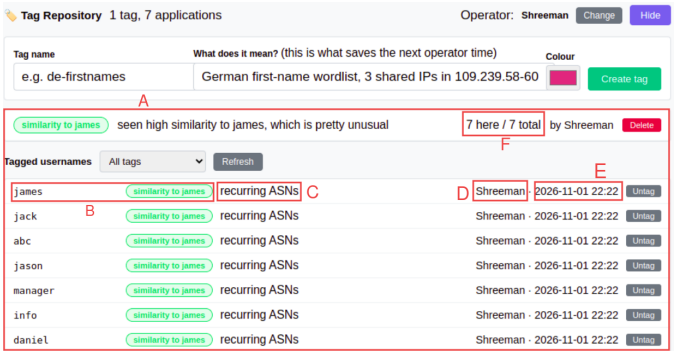


Fig. 8. The tag repository after the tag is applied (A), listing each tagged username (B) with its note (C), author (D), timestamp (E), and application counts (F).

C), the operator who applied it (label D), and a timestamp (label E); any application can be removed with Untag. Tagged usernames also appear in the Tags column of subsequent similarity searches, so an operator who later retrieves them sees immediately that they were already investigated.

Separating a tag's meaning from the rationale for each application lets a later operator understand both what a group represents and why each member belongs to it, without repeating the analysis. The recorded author and timestamp indicate whether a finding is recent. Figs. 6–8 thus show how tagging supports collaboration among operators, fulfilling R3.

V. Evaluation

We evaluate SSHVisual's ability to uncover patterns and show hidden entities in two ways: 1) Security analysts finding campaigns in which the same usernames recur across IPs in different countries (Section V-A) and 2) Network administrators finding entities to block that act alike but are buried underneath others (Section V-B). Both subsections start with an operator finding an IP, e.g., from an intrusion detection alert, an abuse report, or the discovery tables in SSHVisual. On its own, such an IP is one address among many; SSHVisual helps the operator find what else it is connected to.

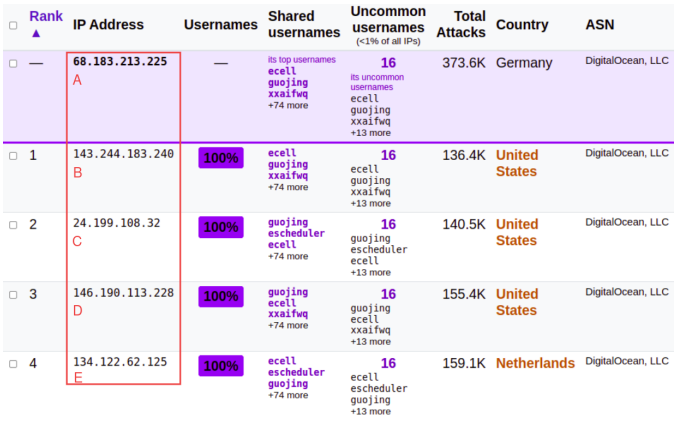
A. Username Campaigns

Fig. 9 shows the username search for IP 68.183.213.225 (DigitalOcean, Germany). Four IPs have 100% username similarity with it (labels A–E): 143.244.183.240, 24.199.108.32, and 146.190.113.228 in the United States, and 134.122.62.125 in the Netherlands. All five try the same 77 usernames in nearly identical proportions, and every list contains all 16 of the searched IP's uncommon usernames, such as ecell, guojing, and xxaifwq. These uncommon usernames make the match meaningful. A 100% score built on common usernames like root and admin would say little, but 16 shared uncommon usernames are unlikely by chance. The five IPs share one ASN but span three countries, which is consistent with a single operator renting servers in several regions. The campaign also extends further: at least 12 IPs score above 84% similarity, spanning five countries and six ASNs (Appendix, Section VI-B).

Fig. 10 shows why this campaign can be hard to see from volume alone. Each of the IPs is active for only a day or two, and at different times: 146.190.113.228 on Dec 19 (label D), 143.244.183.240 on Dec 20 (label B), 24.199.108.32 on Dec 28 (label C), 68.183.213.225 on Jan 2 and Jan 8 (label A), and 134.122.62.125 on Jan 5 to 6 (label E). The lines never rise together, so the chart can suggest five unrelated bursts. The shared username list can link them into one campaign.

B. Finding Hidden Entities to Block

Fig. 11 shows the IP similarity search for 200.195.169.59, an IP from Ligga Telecomunicaç oes S.A. in Brazil with thirty six thousand attacks. Its five closest matches (88.6% to 94.9% behavioral similarity) come from Brazil, Indonesia, Italy, Mexico, and Australia, each from a different ASN. Despite this spread, they behave alike. Each made roughly thirty to forty thousand attacks and is tagged as exploratory targeting, meaning they try many different usernames. Each also shares 1,684 to 1,899 usernames with the searched IP, 46 to 140 of them uncommon (tried by fewer than 1% of all IPs). The searched IP itself tried 3,821 usernames, 925 of them uncommon. The pattern extends beyond these five: the top 20 matches, all at least 86% similar, span 15 countries and 21 ASNs (Appendix, Section VI-C). Together they show



Rank	IP Address	Usernames	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	68.183.213.225 A	—	its top usernames ecell guojing xxaifwq +74 more	16 its uncommon usernames ecell guojing xxaifwq +13 more	373.6K	Germany	DigitalOcean, LLC
1	143.244.183.240 B	100%	ecell guojing xxaifwq +74 more	16 ecell guojing xxaifwq +13 more	136.4K	United States	DigitalOcean, LLC
2	24.199.108.32 C	100%	guojing escheduler ecell +74 more	16 guojing escheduler ecell +13 more	140.5K	United States	DigitalOcean, LLC
3	146.190.113.228 D	100%	guojing ecell xxaifwq +74 more	16 guojing ecell xxaifwq +13 more	155.4K	United States	DigitalOcean, LLC
4	134.122.62.125 E	100%	ecell escheduler guojing +74 more	16 ecell escheduler guojing +13 more	159.1K	Netherlands	DigitalOcean, LLC

Fig. 9. 100% usernames shared among five IPs. IPs A, B, C, D, E match the IPs A, B, C, D, E in Fig. 10.

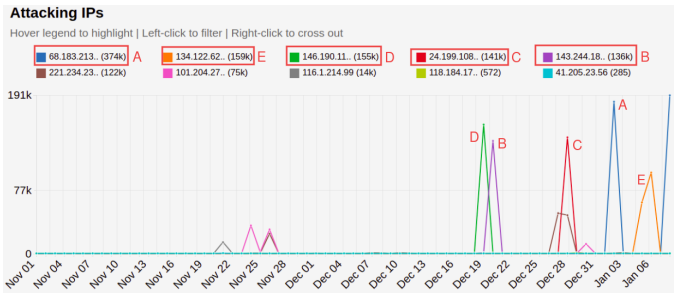


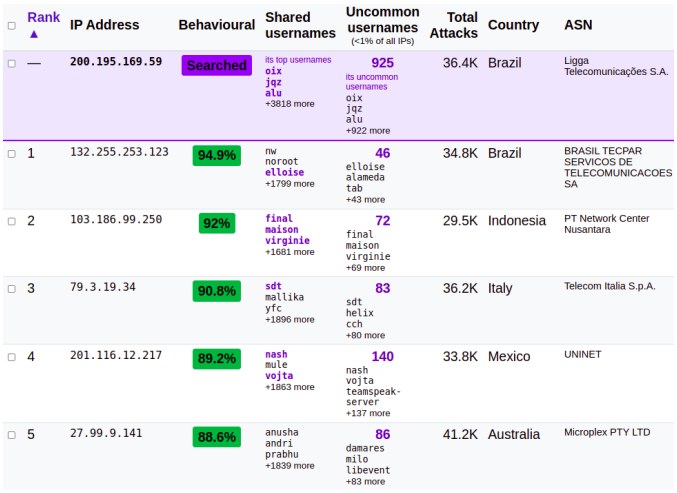
Fig. 10. Five IPs labeled A–E from Fig. 9 in the linechart, along with other IPs with almost the same username list.

low-volume IPs in many countries working through large, overlapping lists of uncommon usernames at the same time.

Fig. 12 shows how buried these IPs are despite their interesting nature to administrators. None of the 10 IPs (selected from Fig. 11) would appear in the top volume IPs chart. The searched IP ranks #296 of 86,985 IPs and accounts for 0.018% of 206.6M attacks. Its country, Brazil, is large enough to appear in the top 10 countries (#8), but its ASN is 13 times too small to appear in the top 10 ASNs, and the IP itself is 20 times too small to appear in the top 10 IPs. The other IPs rank between #222 and #2,826. An administrator looking only at the high volume charts would see Brazil but never these IPs. With SSHVisual, administrators know not just the IP they found to block but also similar IPs that they should block, and most importantly, why they should block them.

VI. Conclusion

We presented SSHVisual, a visual analytics system that treats the username as a first-class analytic dimension alongside the IP, and ASN in SSH attack data. SSHVisual pairs a discovery table that ranks every entity with coordinated linecharts that follow Shneiderman's mantra, so operators can move between precise lookup and temporal behavior. Similarity search groups attackers by behavior along the username and IP dimensions and explains each match. Shared tagging preserves those groups so that findings persist and



Rank	IP Address	Behavioural	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	200.195.169.59	Searched	its top usernames oix jqz alu +3818 more	925 its uncommon usernames oix jqz alu +922 more	36.4K	Brazil	Ligga Telecomunicações S.A.
1	132.255.253.123	94.9%	nw noroot elloise +1799 more	46 elloise alameda tab +43 more	34.8K	Brazil	BRASIL TECPAR SERVICOS DE TELECOMUNICACOES SA
2	103.186.99.250	92%	final maison virginie +1681 more	72 final maison virginie +69 more	29.5K	Indonesia	PT Network Center Nusantara
3	79.3.19.34	90.8%	sdt mallika yfc +1896 more	83 sdt helix cch +80 more	36.2K	Italy	Telecom Italia S.p.A.
4	201.116.12.217	89.2%	nash mule vojta +1863 more	140 nash vojta teamspeak-server +137 more	33.8K	Mexico	UNINET
5	27.99.9.141	88.6%	anusha andri prabhu +1839 more	86 damares milo libevent +83 more	41.2K	Australia	Microplex PTY LTD

Fig. 11. Five IPs that behave like 200.195.169.59.

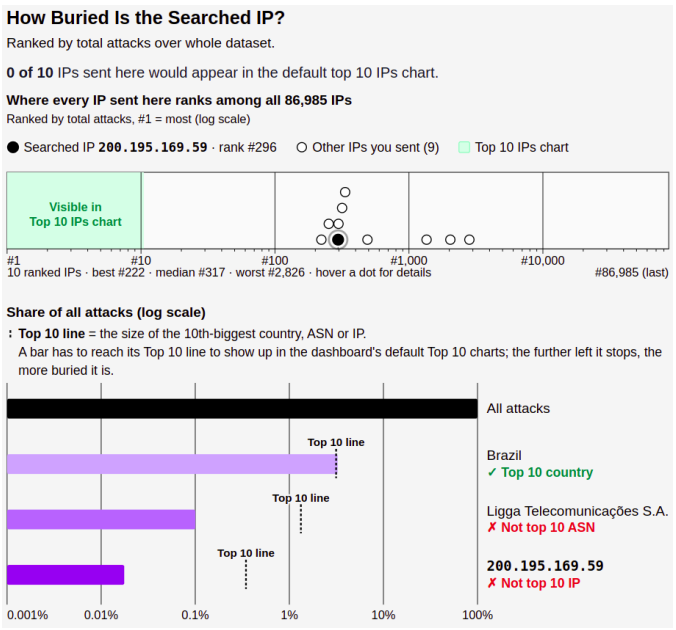


Fig. 12. How buried 200.195.169.59 and its similar IPs are.

remain visible across operators. Our case studies show that this combination surfaces patterns that volume-based, IP-centric views can miss. Low-volume IPs can spread across many countries and ASNs worked through similar lists of uncommon usernames, yet none ranked high enough to appear in any top-ten chart. IPs that never attacked on the same day were linked into a single campaign only by the 77 usernames they shared. Because SSH traffic may not offer a normal baseline for automated detection, these results suggest that exploratory, operator-driven analysis remains a necessary complement to automated methods, and that the username is a valuable lens for recognizing coordinated attackers.

REFERENCES

[1] T. Ylonen, "Ssh–secure login connections over the internet," in *Proceedings of the 6th USENIX Security Symposium*, vol. 37, 1996, pp. 40–52.

[2] S. K. Singh, S. Gautam, C. Cartier, S. Patil, and R. Ricci, "Where the wild things are: Brute-Force SSH attacks in the wild and how to stop them," in *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI 24)*. Santa Clara, CA: USENIX Association, Apr. 2024, pp. 1731–1750. [Online]. Available: https://www.usenix.org/conference/nsdi24/presentation/singh-sachin

[3] M. Antonakakis, T. April, M. Bailey, M. Bernhard, E. Bursztein, J. Cochran, Z. Durumeric, J. A. Halderman, L. Invernizzi, M. Kallitsis *et al.*, "Understanding the mirai botnet," in *26th USENIX security symposium (USENIX Security 17)*, 2017, pp. 1093–1110.

[4] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "Ddos in the iot: Mirai and other botnets," *computer*, vol. 50, no. 7, pp. 80–84, 2017.

[5] F. Dang, Z. Li, Y. Liu, E. Zhai, Q. A. Chen, T. Xu, Y. Chen, and J. Yang, "Understanding fileless attacks on linux-based iot devices with honeycloud," in *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services*, 2019, pp. 482–493.

[6] S. Pastrana and G. Suarez-Tangil, "A first look at the crypto-mining malware ecosystem: A decade of unrestricted wealth," in *Proceedings of the Internet Measurement Conference*, 2019, pp. 73–86.

[7] V. T. Guimarães, C. M. D. S. Freitas, R. Sadre, L. M. R. Tarouco, and L. Z. Granville, "A survey on information visualization for network and service management," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 285–323, 2016.

[8] H. Shiravi, A. Shiravi, and A. A. Ghorbani, "A survey of visualization systems for network security," *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 8, pp. 1313–1329, 2012.

[9] T. Zhang, X. Wang, Z. Li, F. Guo, Y. Ma, and W. Chen, "A survey of network anomaly visualization," *Science China Information Sciences*, vol. 60, no. 12, p. 121101, 2017.

[10] J. Raynor, T. Crnovrsanin, S. Di Bartolomeo, L. South, D. Saffo, and C. Dunne, "The state of the art in bgp visualization tools: A mapping of visualization techniques to cyberattack types," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 1, pp. 1059–1069, 2022.

[11] C. Wu, S. Sheng, and X. Dong, "Research on visualization systems for ddos attack detection," in *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2018, pp. 2986–2991.

[12] C. Gates and C. Taylor, "Challenging the anomaly detection paradigm: A provocative discussion," in *Proceedings of the 2006 workshop on New security paradigms*, 2006, pp. 21–29.

[13] F. Mansmann and S. Mansmann, "Interactive exploration of data traffic with hierarchical network maps," 2006.

[14] J. Oberheide, M. Goff, and M. Karir, "Flamingo: Visualizing internet traffic," in *2006 IEEE/IFIP Network Operations and Management Symposium NOMS 2006*. IEEE, 2006, pp. 150–161.

[15] F. Mansmann, F. Fischer, D. A. Keim, and S. C. North, "Visual support for analyzing network traffic and intrusion detection events using treemap and graph representations," in *Proceedings of the Symposium on Computer Human Interaction for the Management of Information Technology*, 2009, pp. 19–28.

[16] T. Taylor, D. Paterson, J. Glanfield, C. Gates, S. Brooks, and J. McHugh, "Flovis: Flow visualization system," in *2009 Cybersecurity Applications & Technology Conference for Homeland Security*. IEEE, 2009, pp. 186–198.

[17] S. Chen, C. Guo, X. Yuan, F. Merkle, H. Schaefer, and T. Ertl, "Oceans: Online collaborative explorative analysis on network security," in *Proceedings of the Eleventh Workshop on Visualization for Cyber Security*, 2014, pp. 1–8.

[18] T. Tremel, J. Kogel, F. Jauernig, S. Meier, D. Thom, F. Becker, C. Muller, and S. Koch, "Vitalflow: Visual interactive traffic analysis with netflow," in *NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2022, pp. 1–6.

[19] L. Xiao, J. Gerth, and P. Hanrahan, "Enhancing visual analysis of network traffic using a knowledge representation," in *2006 IEEE symposium on visual analytics science and technology*. IEEE, 2006, pp. 107–114.

[20] J. R. Goodall and M. Sowul, "Viassist: Visual analytics for cyber defense," in *2009 IEEE conference on technologies for homeland security*. IEEE, 2009, pp. 143–150.

[21] D. H. Jeong, J.-H. Cho, F. Chen, L. Kaplan, A. Jøsang, and S.-Y. Ji, "Interactive web-based visual analysis on network traffic data," *Information*, vol. 14, no. 1, p. 16, 2022.

[22] H. Choi, H. Lee, and H. Kim, "Fast detection and visualization of network attacks on parallel coordinates," *computers & security*, vol. 28, no. 5, pp. 276–288, 2009.

[23] Y. Livnat, J. Agutter, S. Moon, and S. Foresti, "Visual correlation for situational awareness," in *IEEE Symposium on Information Visualization, 2005. INFOVIS 2005.* IEEE, 2005, pp. 95–102.

[24] C. Humphries, N. Prigent, C. Bidan, and F. Majorczyk, "Elvis: Extensible log visualization," in *Proceedings of the Tenth Workshop on Visualization for Cyber Security*, 2013, pp. 9–16.

[25] T. Yu, R. Lippmann, J. Riordan, and S. Boyer, "Ember: a global perspective on extreme malicious behavior," in *Proceedings of the seventh international symposium on visualization for cyber security*, 2010, pp. 1–12.

[26] D. H. Jeong, B.-K. Jeong, and S.-Y. Ji, "Multi-resolution analysis with visualization to determine network attack patterns," *Applied Sciences*, vol. 13, no. 6, p. 3792, 2023.

[27] A. King, B. Huffaker, A. Dainotti, and K. Claffy, "A coordinated view of the temporal evolution of large-scale internet events," *Computing*, vol. 96, no. 1, pp. 53–65, 2014.

[28] D. Ceneda, M. Di Bartolomeo, V. Di Donato, M. Patrignani, M. Pizzonia, and M. Rimondini, "Routingwatch: Visual exploration and analysis of routing events," in *NOMS 2016-2016 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2016, pp. 591–597.

[29] J. C. Roberts, "State of the art: Coordinated & multiple views in exploratory visualization," in *Fifth international conference on coordinated and multiple views in exploratory visualization (CMV 2007)*. IEEE, 2007, pp. 61–71.

[30] J. S. Yi, Y.-a. Kang, J. T. Stasko, and J. A. Jacko, "Understanding and characterizing insights: how do people gain insights using information visualization?" in *Proceedings of the 2008 Workshop on BEyond time and errors: novel evaLuation methods for Information Visualization*, 2008, pp. 1–6.

[31] R. Djap, C. Lim, K. E. Silaen, and A. Yusuf, "Xb-pot: Revealing honeypot-based attacker's behaviors," in *2021 9th International Conference on Information and Communication Technology (ICoICT)*. IEEE, 2021, pp. 550–555.

[32] V. Visoottiviseth, U. Jaralrungroj, E. Phoomrungraungsuk, and P. Kultanon, "Distributed honeypot log management and visualization of attacker geographical distribution," in *2011 Eighth International Joint Conference on Computer Science and Software Engineering (JCSSE)*. IEEE, 2011, pp. 23–28.

[33] M. Baykara and R. Das, "A novel honeypot based security approach for real-time intrusion detection and prevention systems," *Journal of Information Security and Applications*, vol. 41, pp. 103–116, 2018.

[34] A. Belqruch and A. Maach, "Scada security using ssh honeypot," in *Proceedings of the 2nd International Conference on Networking, Information Systems & Security*, 2019, pp. 1–5.

[35] I. Koniaris, G. Papadimitriou, and P. Nicopolitidis, "Analysis and visualization of ssh attacks using honeypots," in *Eurocon 2013*. IEEE, 2013, pp. 65–72.

[36] A. Kyriakou and N. Sklavos, "Container-based honeypot deployment for the analysis of malicious activity," in *2018 Global Information Infrastructure and Networking Symposium (GIIS)*. IEEE, 2018, pp. 1–4.

[37] I. Koniaris, G. Papadimitriou, P. Nicopolitidis, and M. Obaidat, "Honeypots deployment for the analysis and visualization of malware activity and malicious connections," in *2014 IEEE international conference on communications (ICC)*. IEEE, 2014, pp. 1819–1824.

[38] D. Fraunholz, M. Zimmermann, S. D. Anton, J. Schneider, and H. D. Schotten, "Distributed and highly-scalable wan network attack sensing and sophisticated analysing framework based on honeypot technology," in *2017 7th International Conference on Cloud Computing, Data Science & Engineering-Confluence*. IEEE, 2017, pp. 416–421.

[39] J. R. Goodall, E. D. Ragan, C. A. Steed, J. W. Reed, G. D. Richardson, K. M. Huffer, R. A. Bridges, and J. A. Laska, "Situ: Identifying and explaining suspicious behavior in networks," *IEEE transactions on visualization and computer graphics*, vol. 25, no. 1, pp. 204–214, 2018.

[40] Q. Liao and A. Striegel, "Intelligent network management using graph differential anomaly visualization," in *2012 IEEE Network Operations and Management Symposium*. IEEE, 2012, pp. 1008–1014.

[41] W. A. Pike, C. Scherrer, and S. Zabriskie, "Putting security in context: Visual correlation of network activity with real-world information," in *VizSEC 2007: Proceedings of the Workshop on Visualization for Computer Security*. Springer, 2008, pp. 203–220.
[42] L. Hao, C. G. Healey, and S. E. Hutchinson, "Ensemble visualization for cyber situation awareness of network security data," in *2015 IEEE Symposium on Visualization for Cyber Security (VizSec)*. IEEE, 2015, pp. 1–8.
[43] F. Pouget, G. Urvoy-Keller, and M. Dacier, "Time signatures to detect multi-headed stealthy attack tools," in *18th Annual FIRST Conference*, Baltimore, MD, USA, 2006.
[44] O. Thonnard and M. Dacier, "A framework for attack patterns' discovery in honeynet data," *Digital Investigation*, vol. 5, pp. S128–S139, 2008, proc. DFRWS 2008.
[45] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *2010 IEEE symposium on security and privacy*. IEEE, 2010, pp. 305–316.
[46] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Transactions on Information and System Security (TISSEC)*, vol. 3, no. 3, pp. 186–205, 2000.
[47] D. M. Best, R. P. Hafen, B. K. Olsen, and W. A. Pike, "Atypical behavior identification in large-scale network traffic," in *2011 IEEE Symposium on Large Data Analysis and Visualization*. IEEE, 2011, pp. 15–22.
[48] Y. Xie, F. Yu, K. Achan, E. Gillum, M. Goldszmidt, and T. Wobber, "How dynamic are ip addresses?" in *Proceedings of the 2007 conference on Applications, technologies, architectures, and protocols for computer communications*, 2007, pp. 301–312.
[49] B. Stone-Gross, M. Cova, L. Cavallaro, B. Gilbert, M. Szydlowski, R. Kemmerer, C. Kruegel, and G. Vigna, "Your botnet is my botnet: analysis of a botnet takeover," in *Proceedings of the 16th ACM conference on Computer and communications security*, 2009, pp. 635–647.
[50] G. C. Moura, C. Ganán, Q. Lone, P. Poursaied, H. Asghari, and M. van Eeten, "How dynamic is the isps address space? towards internet-wide dhcp churn estimation," in *2015 IFIP Networking Conference (IFIP Networking)*. IEEE, 2015, pp. 1–9.
[51] J. L. Guerra, E. Veas, and C. A. Catania, "A study on labeling network hostile behavior with intelligent interactive tools," in *2019 IEEE Symposium on Visualization for Cyber Security (VizSec)*. IEEE, 2019, pp. 1–10.
[52] J. Owens and J. Matthews, "A study of passwords and methods used in brute-force SSH attacks," in *USENIX Workshop on Large-Scale Exploits and Emergent Threats (LEET)*, San Francisco, CA, USA, 2008. [Online]. Available: https://lin-web.clarkson.edu/~jmatthew/publications/leet08.pdf
[53] Y. Wu, P. M. Cao, A. Withers, Z. T. Kalbarczyk, and R. K. Iyer, "Mining threat intelligence from billion-scale SSH brute-force attacks," in *Workshop on Decentralized IoT Systems and Security (DISS)*. San Diego, CA, USA: Internet Society, 2020. [Online]. Available: https://www.ndss-symposium.org/wp-content/uploads/2020/04/diss2020-23007-paper.pdf
[54] B. Shneiderman, "The eyes have it: A task by data type taxonomy for information visualizations," in *Proceedings of the IEEE Symposium on Visual Languages*, 1996, pp. 336–343.
[55] W. Javed, B. McDonnel, and N. Elmqvist, "Graphical perception of multiple time series," *IEEE Transactions on Visualization and Computer Graphics*, vol. 16, no. 6, pp. 927–934, 2010.

APPENDIX

A. Discovery Table Columns

Table I lists every column of the discovery table (Section IV-A1) and the tabs in which it appears.

TABLE I
DISCOVERY TABLE COLUMNS BY TAB

Column	Countries	IPs	ASNs	Usernames
Total attacks	✓	✓	✓	✓
Average daily attacks	✓	✓	✓	✓
Persistence	✓	✓	✓	✓
Max absolute Δ	✓	✓	✓	✓
Max % Δ	✓	✓	✓	✓
Attacks in last 7 days	✓	✓	✓	✓
First / last seen	✓	✓	✓	✓
Max daily count	✓	✓	✓	✓
Trend	✓	✓	✓	✓
Burst intensity	✓	✓	✓	✓
Unique usernames	✓	✓	✓	
Username stability	✓	✓	✓	
Username concentration	✓	✓	✓	
Unique IPs	✓		✓	✓
IP stability	✓		✓	✓
IP concentration	✓		✓	✓
Unique ASNs	✓			✓
ASN stability	✓			✓
ASN concentration	✓			✓
Unique countries			✓	✓
Country stability			✓	✓
Country concentration			✓	✓
ASN name, Country		✓		
Primary country			✓	
Behavior badge	✓			

B. Username Campaigns

C. Hidden Entities